

Moneyball your MLB Pitching Staff across Roles with FIP_Pyth_WinRate and FIP_Pyth_WL: ML-Validated, All-Role Scalable Measures of Pitcher Win Output per Opportunity (Arbitrage) & Pitching's Share of Winning

• Baseball Track

Introduction. Pitcher evaluation has advanced from win-loss record, or saves and holds in relief, through earned run average (ERA), Fielding Independent Pitching (FIP), and pitcher Wins Above Replacement (pWAR). This progression sacrificed the team-aligned output unit per opportunity. No existing measure provides a win-rate (loss-rate) *per pitching opportunity*, and relief pitchers were never evaluated in an apples-to-apples framework. Win-Loss, Saves, and Holds are 3 different measures with distinct units, none being high-signal or team-aligned. Then, the founding *Moneyball* question (*Which inputs does the market misprice?*) has never been answerable across a pitching-staff. Even *pWAR* does not balance a pitcher's opportunities with their success-rate. **ML ridge-regression** shows that a 2-WAR middle-reliever (closer, starter) supports a win-rate of 0.589 (0.615, 0.511), with IP-differential being the confound. Using tools from ML and game theory, we develop **FIP_Pyth_WinRate** and **FIP_Pyth_WL**, which map pitcher-*FIP*, league-average *FIP*, and league-average unearned run-rate to an MLB-optimized Pythagorean win-function (contest success function in game theory) to express any pitcher's output as an advanced expected win-rate per 9-inning opportunity and an advanced win-loss record, in equivalent terms for all pitching-roles. The measures also meritoriously divide/assign partial-wins to pitchers appearing in a game without confounds like game-innings pitched, run-support, or starter-designation.

Methods. We estimate the following equations and fit them onto cross-role pitcher win-output and marginal-revenue product 3-D surfaces.

$$W\%_{team} = \frac{RS^\gamma}{RS^\gamma + RA^\gamma}, \quad w_p = \frac{(F^* + u)^\gamma}{(F^* + u)^\gamma + (FIP_p + u)^\gamma}, \quad (W_p, L_p) = (G_p w_p, G_p(1 - w_p)) \quad (1)$$

$W\%_{team}$ is team Pythagorean expected-win%; w_p is **FIP_Pyth_WinRate** and (W_p, L_p) is **FIP_Pyth_WL**, RS and RA are runs scored and allowed per game, $\gamma = 1.83$, FIP_p is pitcher p 's *FIP*, u the league unearned-run rate, F^* the win-neutral *FIP* at which a staff breaks even, and $G_p = IP_p/9$ their nine-inning opportunities. The first equation in (1) estimates γ from **nonlinear Weibull regression**. The second maps the optimal Pythagorean-wins function to the individual-pitcher *FIP*-level. It asks the pitcher's win-rate per 9-innings versus league-average opposing-*FIP* and a league-average unearned run-rate across teams. The third equation scales this win (loss) rate by full-game equivalents pitched to develop a valid W-L record for any pitcher.

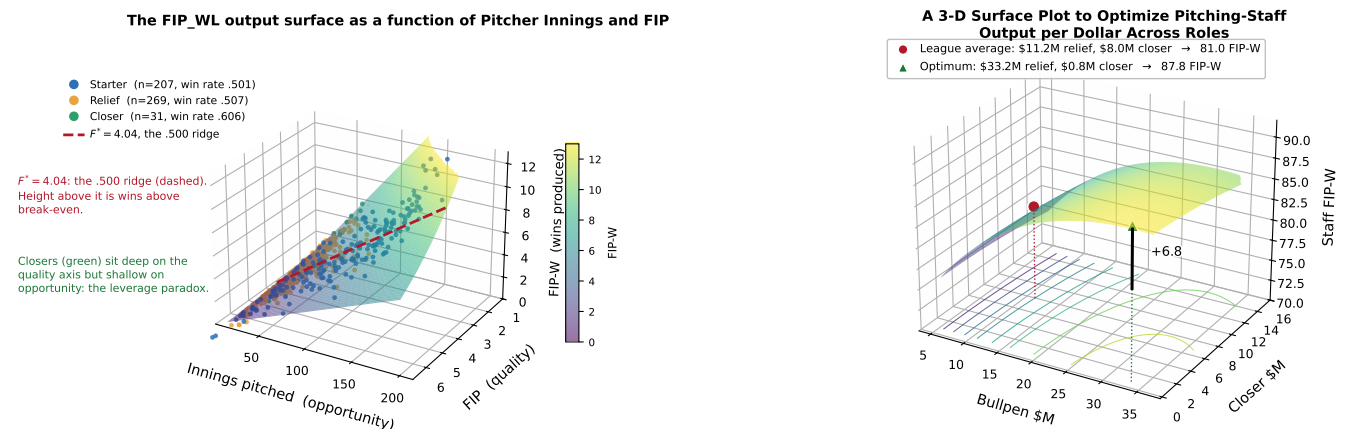


Figure 1: Left: every 2026 pitcher on a single surface — quality against opportunity, elevation in wins — with the dashed red ridge marking the win-neutral anchor at which a pitcher is exactly break-even. Starters, relievers and closers are graded on the same curve. **Right:** staff wins over bullpen and closer spending on a fixed \$53.2M payroll, the rotation absorbing the residual; the club climbs from 81–81 to 88–74.

Results. Estimating on complete *Baseball Reference* MLB pitcher-data for 2023—2026, we validate and apply the measures using ML and statistical-learning models. We **ridge regress** team-season wins on **FIP_Pyth_WL**-estimated wins, runs the defense saved or surrendered, and runs the offense scored. Whereas direct *FIP*-measures

fail to aggregate to a 0.5 implied league-average win-rate by **Jensen's inequality**, `FIP_Pyth_WinRate` does given its optimized-Pythagorean win-calibration. An MLB team-season decomposes into five win-inputs explaining **98.8% of variation in winning: offense at 37.9%**, sequencing of baserunners and outs at **26.3%**, **FIP-wins at 20.9%**, fate of balls in play at 12.3%, and fielding errors at 1.4%. Chance sequencing, or random-clustering of on-base events, changes ERA. It is sequencing, not fielding, that ERA mostly mistakes for pitching. `FIP_Pyth_WinRate` and `FIP_Pyth_WL` isolate fielding, sequencing, and pitching. Pitching, at 20.9%, is significantly more win-productive than fielding (e.g., in **bootstrap resamples**).

Ridge, random forest, and gradient boosting train on the 2023-to-2024 and 2024-to-2025 transitions and test on the 2025-to-2026 transition. `FIP_Pyth_WinRate` is **twice as persistent across season as ERA**, demonstrating a substantial signal-to-noise/skill-to-noise gain. `FIP_Pyth_WinRate` permutation importance on **held-out seasons** ranks first against twelve rival-inputs, **4 times stronger than ERA**, in picking best-available pitchers of 2025-26 blind. **Conformal quantile-boosting** gives each pitcher a projection-range, so signing is judged on downside, midpoint, and upside. Lastly, **decision-weighted boosting** fits wins on runs-allowed and confirms the team-optimized Pythagorean exponent holds at individual pitcher-level.

Conclusion. After calibrating/validating of the measures as pushing the MLB analytic-frontier, we apply them for the first known apples-to-apples *Moneyball* analysis of the whole pitching-staff. Merging 819 individual 2026 contracts, closers earn a median \$7.50M against \$0.80M for other relievers on fewer innings, **\$21M per marginal win**. Crossing role with service class, a win costs \$0.13M from a pre-arbitration starter, \$0.39M from a pre-arbitration reliever, \$0.90M from a veteran reliever, \$3.16M from a veteran starter, and \$3.32M from a veteran closer—a **twenty-six-fold spread** in which class dominates role and the role effect reverses.

We conclude that a league-average team can go from 81-81 to 88-74 in expectation without spending any more money, by changing who fills three sets of roster spots. First, the team should not re-sign its closer, and instead give the role to the best-available pre-arbitration reliever at a \$7⁺M savings. Second, the team should replace its fourth and fifth starters with cheaper ones, cutting average pay for those two spots from \$6.80M to \$3.85M. Third, the team spends everything it freed up, roughly \$22M, on its seven middle-relief spots, raising pay per spot from \$1.60M to \$4.74M. The bullpen throws the same number innings with better pitchers. On the same \$53.2M payroll, the record implied by the pitching improves from **81–81 to 88–74**. The strategy is win-positive in 80% of **200,000** simulated seasons, and 92.7% cumulatively if run for three winters, as the gain accumulates linearly while its dispersion grows with the square root of seasons.

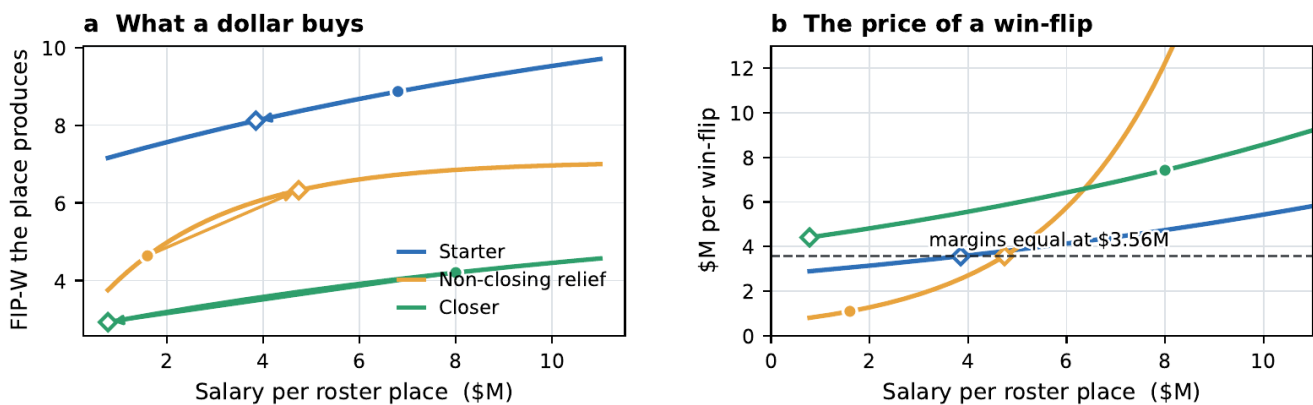


Figure 2: (a) What a dollar buys at each role: the concave gradients fitted to individual 2026 contracts. (b) The resulting marginal cost of flipping one loss into one win — \$1.09M in the bullpen, \$4.36M in the rotation, \$7.42M at closer — with a club buying until the three margins meet. Circles mark the league-average contract at each role, diamonds the reallocated position.